# EMPS Multi-area power market simulator

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https://www.sintef.no/en/software/emps-multi-area-power-market-simulator/

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Software

EMPS - multi area power-market simulator

EMPS (EFI's Multi-area Power-market Simulator) a software program for the optimization and simulation of hydrothermal power systems. The model accounts for transmission constraints and hydrological differences between subregions.

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In deregulated markets, the model has become widely used for forecasting future spot prices. It is also used for investment analyses, for example for new transmission lines or power plants. In addition to forecasts of future spot prices, the model provides forecasts for generation, reservoir levels, water inflow, consumption, rationing, imports/exports, and more. The model is used by most major players in the Nordic power market.

The model is continuously developed to adapt to changes in the power system and has recently been enhanced with improved functionality for modelling short-term flexibility, including batteries, and flow-based market coupling.

The computational methodology used in EMPS is highly efficient, enabling the model to find high-quality solutions significantly faster than alternative solution methods.

Vansimtap and EMPS - multi area power - market simulator

These are among the most widely used energy models in the Nordic power market. They are used for generation planning (long- and medium-term), expansion planning, and power system and market analysis. The models are best suited for systems with a significant share of hydropower.

The models can be run as standalone applications or accessed through Python functions.

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